APPLIED ADVANCED OPTIMISATION

1. a) i) The problem can be formulated as

$$\begin{aligned} \min_{x} \quad & \sum_{i=1}^{N} \frac{1}{C_i - x_i} + \lambda \sum_{i=1}^{N} (\alpha_i x_i^2 + \beta_i x_i) \\ \text{s.t.} \quad & \sum_{i=1}^{N} x_i = L \\ & 0 \le x_i < C_i, \quad i = 1, \ldots, N \end{aligned}$$

2 marks to the objective function, 1 mark for each constraint (three constraints). [5 marks]

ii) Let $f(x) = (C-x)^{-1}$. The first derivative is $f'(x) = -(C-x)^{-2}(-1) = (C-x)^{-2}$. The second derivative is:

$$f''(x) = -2(C-x)^{-3}(-1) = \frac{2}{(C-x)^3}$$

Since the domain requires $x < C$, the term $(C-x)$ is positive, implying $(C-x)^3 > 0$. Therefore, $f''(x) > 0$ for all x in the domain. A function with a non-negative second derivative is convex. [2 marks]

iii) The parameter λ controls the trade-off between the two competing objectives (latency and power).

- By varying $\lambda > 0$, we trace out the Pareto optimal frontier.
- The solution $x^*(\lambda)$ represents a specific Pareto optimal point where the marginal reduction in latency equals λ times the marginal increase in power cost. Large λ prioritises power saving; small λ prioritises latency reduction.

1 mark for each of the two observations. [2 marks]

iv) Standard convex optimisation problems are typically defined over closed sets (using $\le$) to ensure that the minimum is attained. A strict inequality ($x_i < C_i$) defines an open set. Note that in this specific case the objective function approaches infinity as $x_i \to C_i$ (barrier property) which forces the solution to be in the interior. [1 mark]

b) i) The cost function is $3(\pi r^2) + 1(\pi r^2) + 1(2\pi rh) = 4\pi r^2 + 2\pi rh$. The problem in posynomial form is

$$\begin{aligned} \min_{r,h} \quad & 4\pi r^2 + 2\pi rh \\ \text{s.t.} \quad & V_{min} \pi^{-1} r^{-2} h^{-1} \le 1 \\ & 2rh^{-1} \le 1 \\ & 0.25hr^{-1} \le 1 \\ & L_{max}^{-1} h + 2L_{max}^{-1} r \le 1 \end{aligned}$$

1 mark for the objective function, 1 mark for each constraint (four constraints). [5 marks]

Answers

ii) Let $r = e^{y_r}$ and $h = e^{y_h}$. Taking the log of the objective and constraints:

$$\begin{aligned} \min_{y_r, y_h} \quad & \log(4\pi e^{2y_r} + 2\pi e^{y_r + y_h}) \\ \text{s.t.} \quad & \log(V_{min}\pi^{-1}) - 2y_r - y_h \leq 0 \\ & \log(2) + y_r - y_h \leq 0 \\ & \log(0.25) + y_h - y_r \leq 0 \\ & \log(e^{y_h} + 2e^{y_r}) - \log(L_{max}) \leq 0 \end{aligned}$$

1 mark for the objective function, 1 mark for each constraint (four constraints). [5 marks]

Answers

2. a) The conjugate of a function $f(x)$ is defined as $f^*(y) = \sup_{x \in \text{dom } f} (yx - f(x))$. Substituting $f(x) = x\log x$:

$$f^*(y) = \sup_{x>0}(yx - x\log x)$$

To find the supremum, we define $h(x) = yx - x\log x$ and find the stationary point by setting the derivative to zero

$$h'(x) = y - (\log x + 1) = 0$$

$$\implies \log x = y - 1$$

$$\implies x^* = e^{y-1}$$

Since $h''(x) = -1/x < 0$ for $x > 0$, this point is a maximum. Substituting x^* back into the expression for $f^*(y)$

$$f^*(y) = y(e^{y-1}) - e^{y-1}\log(e^{y-1}),$$

$$f^*(y) = ye^{y-1} - e^{y-1}(y-1),$$

$$f^*(y) = e^{y-1}(y - y + 1) = e^{y-1}.$$

1 mark for writing $f^*(y)$, 1 mark for $h'(x)$, 1 mark for x^*, 1 mark for $f^*(y)$.

[4 marks]

b) The Lagrangian for the problem is

$$L(x, \nu) = \sum_{i=1}^{n} x_i \log x_i + \nu \left(\sum_{i=1}^{n} x_i - 1 \right)$$

Rearranging the terms to separate the sum

$$L(x, \nu) = \sum_{i=1}^{n} (x_i \log x_i + \nu x_i) - \nu$$

The dual function $g(\nu)$ is the infimum of the Lagrangian with respect to x

$$g(\nu) = \inf_x L(x, \nu) = \sum_{i=1}^{n} \inf_{x_i} (x_i \log x_i + \nu x_i) - \nu \tag{2.1}$$

Using the hint, the term $\inf_{x_i}(x_i \log x_i + \nu x_i)$ is equivalent to $-f^*(-\nu)$, where $f(x) = x\log x$ and the linear coefficient is interpreted as $-y = \nu \implies y = -\nu$. Using the result from part (a)

$$\inf_{x_i}(x_i \log x_i + \nu x_i) = -f^*(-\nu) = -e^{-\nu-1}$$

Substituting this back into the sum:

$$g(\nu) = \sum_{i=1}^{n} (-e^{-\nu-1}) - \nu = -ne^{-\nu-1} - \nu.$$

2 marks for $L(x, \nu)$, 2 marks for (2.1)*, 2 marks for final $g(\nu)$.*

[6 marks]

c) The KKT conditions for this problem are:

Answers

- **Primal Feasibility:**

$$\sum_{i=1}^{n} x_i = 1, \quad x_i > 0 \quad \forall i$$

- **Dual Feasibility:** None (since there are no inequality constraints, ν is unconstrained).
- **Complementary Slackness:** None (no inequality constraints).
- **Stationarity:**

$$\nabla_x L(x, \nu) = 0$$

$$\frac{\partial L}{\partial x_i} = \log x_i + 1 + \nu = 0 \quad \forall i = 1, \ldots, n$$

2 marks for primal feasibility, 2 marks for stationarity.

[4 marks]

d) i) From the KKT stationarity condition in part (c):

$$\log x_1 + 1 + \nu = 0 \qquad \log x_2 + 1 + \nu = 0.$$

By subtracting the second equation from the first, we obtain

$$\log x_1 - \log x_2 = \log \frac{x_1}{x_2} = 0 \implies x_1 = x_2.$$

Substituting this into $x_1 + x_2 = 1$ yields $x_1^* = x_2^* = \frac{1}{2}$. The optimal primal value is

$$p^* = \sum_{i=1}^{2} x_i^* \log x_i^* = \frac{1}{2}\log\left(\frac{1}{2}\right) + \frac{1}{2}\log\left(\frac{1}{2}\right) = \log\left(\frac{1}{2}\right) = -\log 2.$$

2 marks for x^*, 1 mark for p^*.

[3 marks]

ii) From part (b) with $n = 2$, the dual function is $g(\nu) = -2e^{-\nu-1} - \nu$. To find the maximum, set $g'(\nu) = 0$:

$$g'(\nu) = -2(-1)e^{-\nu-1} - 1 = 2e^{-\nu-1} - 1 = 0$$

$$2e^{-\nu-1} = 1 \implies e^{-\nu-1} = \frac{1}{2}$$

Taking the log of both sides:

$$-\nu - 1 = \log(1/2) = -\log 2$$

$$\nu^* = \log 2 - 1$$

Now, calculate the optimal dual value $d^* = g(\nu^*)$:

$$d^* = -2e^{-(\log 2 - 1)-1} - (\log 2 - 1)$$

$$d^* = -2e^{-\log 2} - \log 2 + 1$$

$$d^* = -2(1/2) - \log 2 + 1$$

$$d^* = -1 - \log 2 + 1 = -\log 2$$

Since $p^* = -\log 2$ and $d^* = -\log 2$, we have $p^* = d^*$, confirming Strong Duality holds.

1 mark for ν^*, 1 mark for d^*, 1 mark for strong duality.

[3 marks]

Answers

3. a) i) $f(x) = x^T x$**:** The function is $f(x) = ||x||_2^2$. The gradient is $\nabla f(x) = 2x$. The Hessian is $\nabla^2 f(x) = 2I$, where I is the identity matrix. Since the eigenvalues of I are all 1, the eigenvalues of $2I$ are all $2 > 0$. Therefore, the Hessian is positive definite everywhere ($2I \succ 0$), which implies the function is strictly convex.

1 mark for the Hessian, 1 mark for the conclusion. [2 marks]

ii) We calculate the Hessian matrix. The first partial derivatives are

$$\frac{\partial f}{\partial x_1} = \frac{2x_1}{x_2}, \quad \frac{\partial f}{\partial x_2} = -\frac{x_1^2}{x_2^2}.$$

The Hessian is

$$\nabla^2 f(x) = \begin{bmatrix} \frac{2}{x_2} & -\frac{2x_1}{x_2^2} \\ -\frac{2x_1}{x_2^2} & \frac{2x_1^2}{x_2^3} \end{bmatrix} = \frac{2}{x_2^3}\begin{bmatrix} x_2^2 & -x_1x_2 \\ -x_1x_2 & x_1^2 \end{bmatrix}$$

To check for convexity, we examine if $v^T \nabla^2 f(x) v \geq 0$ for any vector $v \in \mathbb{R}^2$:

$$v^T \nabla^2 f(x) v = \frac{2}{x_2^3}(x_2^2 v_1^2 - 2x_1x_2v_1v_2 + x_1^2 v_2^2) = \frac{2}{x_2^3}(x_2v_1 - x_1v_2)^2$$

Since $x_2 > 0$ (domain is $\mathbb{R}_{++}$) and the squared term is non-negative, the quadratic form is always non-negative. Thus, the Hessian is positive semi-definite and the function is convex. Alternatively, one can notice

$$\nabla^2 f(x) = \begin{bmatrix} \frac{2}{x_2} & -\frac{2x_1}{x_2^2} \\ -\frac{2x_1}{x_2^2} & \frac{2x_1^2}{x_2^3} \end{bmatrix} = \frac{2}{x_2}\begin{bmatrix} 1 \\ -\frac{2x_1}{x_2} \end{bmatrix}\begin{bmatrix} 1 & -\frac{2x_1}{x_2} \end{bmatrix}$$

Reaching the same conclusion.

1 mark for the Hessian, 1 mark for the conclusion. [2 marks]

iii) First, we compute the gradient.

$$\frac{\partial f}{\partial x_1} = \frac{e^{x_1}}{e^{x_1} + e^{x_2}}, \quad \frac{\partial f}{\partial x_2} = \frac{e^{x_2}}{e^{x_1} + e^{x_2}}$$

Next, we compute the Hessian matrix

$$\nabla^2 f(x) = \frac{e^{x_1}e^{x_2}}{(e^{x_1} + e^{x_2})^2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

The eigenvalues of the constant matrix are 0 and 2, thus the matrix is positive semi-definite and the function is convex.

1 mark for the Hessian, 1 mark for the conclusion. [2 marks]

iv) Let S_1 and S_2 be convex sets. Let $x, y \in S_1 \cap S_2$. By definition of intersection, $x \in S_1$ and $x \in S_2$, and similarly $y \in S_1$ and $y \in S_2$. Since S_1 is convex, for any $\theta \in [0,1]$, the point $z = \theta x + (1-\theta)y \in S_1$. Since S_2 is convex, for any $\theta \in [0,1]$, the point $z = \theta x + (1-\theta)y \in S_2$. Therefore, $z \in S_1 \cap S_2$. Thus $S_1 \cap S_2$ is a convex set. [2 marks]

Answers

b) We introduce Lagrange multipliers ν for the equality constraints $Ax-b=0$ and λ for the inequality constraints. The Lagrangian is:

$$L(x,\lambda,\nu) = c^\top x + \nu^\top (Ax-b) - \lambda^\top x$$

where $\lambda \succeq 0$. Rearranging terms to isolate x

$$L(x,\lambda,\nu) = (c + A^\top \nu - \lambda)^\top x - b^\top \nu.$$

The dual function is the infimum of the Lagrangian with respect to x:

$$g(\lambda,\nu) = \inf_x L(x,\lambda,\nu) = \begin{cases} -b^\top \nu & \text{if } c + A^\top \nu - \lambda = 0 \\ -\infty & \text{otherwise} \end{cases}$$

The dual problem is to maximise $g(\lambda,\nu)$ subject to $\lambda \succeq 0$. Making the implicit equality constraints explicit, the dual problem is:

$$\begin{array}{ll} \max & -b^\top \nu \\ s.t. & c + A^\top \nu - \lambda = 0 \\ & \lambda \succeq 0 \end{array}$$

Eliminating λ yields the standard form

$$\begin{array}{ll} \max & -b^\top \nu \\ s.t. & c + A^\top \nu \succeq 0 \end{array}$$

1 mark for L, 1 mark for g, 1 mark for the max problem, 1 mark for eliminating λ. [4 marks]

c) i) For $f(x) = \frac{1}{2} x^T P x + q^T x + r$, the gradient is $\nabla f(x) = Px + q$ and the Hessian is $\nabla^2 f(x) = P$. The Newton step Δx_{nt} is defined as

$$\Delta x_{nt} = -(\nabla^2 f(x))^{-1} \nabla f(x),$$

namely

$$\Delta x_{nt} = -P^{-1}(Px+q) = -x - P^{-1} q.$$

1 mark for the gradient, 1 mark for the Hessian, 1 mark for the step.

[3 marks]

ii) The next iterate is:

$$x^{(1)} = x^{(0)} + \Delta x_{nt} = x^{(0)} + (-x^{(0)} - P^{-1} q) = -P^{-1} q$$

We check the optimality condition $\nabla f(x) = 0$ at $x^{(1)}$:

$$\nabla f(x^{(1)}) = P(-P^{-1} q) + q = -q + q = 0.$$

Since P is positive definite, the function is strictly convex, and the point where the gradient vanishes is the unique global minimiser. Thus, the method converges in one step.

1 mark for the next iterate, 1 mark for checking the optimality condition, 1 mark for the global explanation.

[3 marks]

Answers

iii) Gradient descent updates using the direction $-\nabla f(x) = -(Px+q)$. This is the direction of steepest descent, which is perpendicular to the level sets (contours) of f. For a quadratic function, the level sets are ellipsoids defined by the matrix P. If the condition number of P (ratio of largest to smallest eigenvalue) is large, these ellipsoids are highly elongated (narrow valleys). In this geometry, the steepest descent direction rarely points towards the centre (the minimum), causing the algorithm to "zig-zag" across the valley rather than moving efficiently towards the solution. Newton's method corrects for this curvature by using P^{-1}, effectively transforming the ellipsoids into circles where the gradient points directly to the centre.

1 mark for why Gradient Descent does not converge in one step, 1 mark for why Newton does. [2 marks]

Answers